CPM Core Connections Course 1–3 大纲与海报计划

版本：CPM Educational Program（美国加州的非营利组织）出版的 **Core Connections, Courses 1–3**，对应六、七、八年级。教材以小组合作解题为主，**没有开放许可**（© CPM Educational Program，版权所有），本项目只引用公开的目录信息。

目录来源：章名逐字摘自 CPM 官网中学页面的 Table of Contents（Course 1 和 Course 2 各 9 章，Course 3 共 10 章）；每课的编号和英文标题逐字摘自 CPM 官网公开的三份《Text to CCSS Standards》对照表 PDF（各课和课程标准的对应关系）。课的编号可以和 CPM 官方的 Homework Help 网站（homework.cpm.org）对照。中文章名是本项目的翻译；课名保留英文原文，不逐条翻译。

海报粒度：每章一张，全套 28 张（Course 1 九张、Course 2 九张、Course 3 十张），共 256 课。每张海报 3 行，挑这一章最核心的 3 个内容。

本地化：英制单位改为公制，美元改为人民币。CPM 的原题没有公开，海报里的情境和数字都是自编的同类例子，标「示例」。

来源

- 章名：CPM Math · Middle School（Core Connections Table of Contents）
- 课名：Course 1 Text to CCSS（PDF） · Course 2（PDF） · Course 3（PDF）
- 对照：CPM Homework Help、Parent Guide

Course 1（六年级） • 9 章 • 80 课

第1章 导入和表示（Introduction & Representation）

- 来源：Course 1 Text to CCSS Standards（PDF）
- Lesson 1.1.1：Visualizing Information
- Lesson 1.1.2：Perimeter and Area Relationships
- Lesson 1.1.3：Describing and Extending Patterns
- Lesson 1.1.4：Representing Data
- Lesson 1.1.5：Making Sense of a Logic Problem
- Lesson 1.2.1：Multiple Representations
- Lesson 1.2.2：Representing Comparisons
- Lesson 1.2.3：Characteristics of Numbers
- Lesson 1.2.4：Products, Factors, and Factor Pairs

第2章 计算策略和面积（Arithmetic Strategies & Area）

- 来源：Course 1 Text to CCSS Standards（PDF）
- Lesson 2.1.1：Dot Plots and Bar Graphs
- Lesson 2.1.2：Histograms and Stem-and-Leaf Plots
- Lesson 2.2.1：Exploring Area
- Lesson 2.2.2：Square Units and Area of Rectangles
- Lesson 2.2.3：Area and Perimeter
- Lesson 2.3.1：Using Rectangles to Multiply
- Lesson 2.3.2：Using Generic Rectangles
- Lesson 2.3.3：Generic Rectangles and the Greatest Common Factor
- Lesson 2.3.4：Distributive Property

第3章 部分量和整数（Portions & Integers）

- 来源：Course 1 Text to CCSS Standards（PDF）
- Lesson 3.1.1：Using the Multiplicative Identity
- Lesson 3.1.2：Portions as Percents
- Lesson 3.1.3：Connecting Percents with Decimals and Fractions
- Lesson 3.1.4：Multiple Representations of a Portion
- Lesson 3.1.5：Completing the Web
- Lesson 3.1.6：Investigating Ratios
- Lesson 3.2.1：Addition, Subtraction, and Opposites
- Lesson 3.2.2：Locating Negative Numbers
- Lesson 3.2.3：Absolute Value
- Lesson 3.2.4：Length on a Coordinate Graph

第4章 变量和比（Variables & Ratios）

- 来源：Course 1 Text to CCSS Standards（PDF）

- Lesson 4.1.1：Introduction to Variables
- Lesson 4.1.2：Writing Equivalent Expressions
- Lesson 4.1.3：Using Variables to Generalize
- Lesson 4.2.1：Enlarging Two-Dimensional Shapes
- Lesson 4.2.2：Enlarging and Reducing Figures
- Lesson 4.2.3：Enlargement and Reduction Ratios
- Lesson 4.2.4：Ratios in Other Situations

第5章 分数乘法和面积（Multiplying Fractions & Area）

- 来源：Course 1 Text to CCSS Standards（PDF）
- Lesson 5.1.1：Representing Fraction Multiplication
- Lesson 5.1.2：Describing Parts of Parts
- Lesson 5.1.3：Calculating Parts of Parts
- Lesson 5.1.4：Multiplying Mixed Numbers
- Lesson 5.2.1：Making Sense of Decimal Multiplication
- Lesson 5.2.2：Fraction Multiplication Number Sense
- Lesson 5.3.1：Rearranging Areas
- Lesson 5.3.2：Area of a Parallelogram
- Lesson 5.3.3：Area of a Triangle
- Lesson 5.3.4：Area of a Trapezoid

第6章 除法和代数式（Dividing & Building Expressions）

- 来源：Course 1 Text to CCSS Standards（PDF）
- Lesson 6.1.1：Dividing
- Lesson 6.1.2：Fractions as Division Problems
- Lesson 6.1.3：Problem Solving with Division
- Lesson 6.1.4：Solving Problems Involving Fraction Division
- Lesson 6.2.1：Order of Operations
- Lesson 6.2.2：Area of a Rectangular Shape
- Lesson 6.2.3：Naming Perimeters of Algebra Tiles
- Lesson 6.2.4：Combining Like Terms
- Lesson 6.2.5：Evaluating Algebraic Expressions

第7章 比率和运算（Rates & Operations）

- 来源：Course 1 Text to CCSS Standards（PDF）
- Lesson 7.1.1：Comparing Rates
- Lesson 7.1.2：Comparing Rates with Tables and Graphs
- Lesson 7.1.3：Unit Rates
- Lesson 7.2.1：Analyzing Strategies for Dividing Fractions
- Lesson 7.2.2：Another Strategy for Dividing Fractions
- Lesson 7.2.3：Division with Fractions and Decimals
- Lesson 7.2.4：Fraction Division as Ratios
- Lesson 7.3.1：Inverse Operations

- Lesson 7.3.2：Distributive Property
- Lesson 7.3.3：Distributive Property and Expressions Vocabulary
- Lesson 7.3.4：Writing Algebraic Equations and Inequalities

第8章 统计和乘法方程（Statistics & Multiplication Equations）

- 来源：Course 1 Text to CCSS Standards（PDF）
- Lesson 8.1.1：Measures of Central Tendency
- Lesson 8.1.2：Choosing Mean or Median
- Lesson 8.1.3：Shape and Spread
- Lesson 8.1.4：Box Plots and Interquartile Range
- Lesson 8.1.5：Comparing and Choosing Representations
- Lesson 8.2.1：Statistical Questions
- Lesson 8.3.1：Writing Multiplication Equations
- Lesson 8.3.2：Distance, Rate, and Time
- Lesson 8.3.3：Unit Conversion

第9章 体积和百分数（Volume & Percents）

- 来源：Course 1 Text to CCSS Standards（PDF）
- Lesson 9.1.1：Volume of a Rectangular Prism
- Lesson 9.1.2：Nets and Surface Area
- Lesson 9.2.1：Multiplicative Growth and Percents
- Lesson 9.2.2：Composition and Decomposition of Percents
- Lesson 9.2.3：Percent Discounts
- Lesson 9.2.4：Simple Interest and Tips

Course 2（七年级） · 9 章 · 91 课

第1章 导入和概率（Introduction & Probability）

- 来源：Course 2 Text to CCSS Standards（PDF）
- Lesson 1.1.1：Finding Shared and Unique Characteristics
- Lesson 1.1.2：Analyzing a Game
- Lesson 1.1.3：Finding Unknowns
- Lesson 1.1.4：Investigating a Proportional Relationship
- Lesson 1.1.5：Investigating Number Patterns
- Lesson 1.2.1：Introduction to Probability
- Lesson 1.2.2：Investigating Probability
- Lesson 1.2.3：Modifying the Sample Space
- Lesson 1.2.4：Expressing Fractions as Percents
- Lesson 1.2.5：Rewriting Fractions
- Lesson 1.2.6：Fraction Addition
- Lesson 1.2.7：Compound Probability
- Lesson 1.2.8：Subtracting Probabilities

第2章 分数和整数加法（Fractions & Integer Addition）

- 来源：Course 2 Text to CCSS Standards（PDF）
- Lesson 2.1.1：Fraction-to-Decimal Conversions
- Lesson 2.1.2：Rewriting Decimals as Fractions
- Lesson 2.2.1：Composing Integers
- Lesson 2.2.2：Adding Integers and Rational Numbers
- Lesson 2.2.3：More Addition of Integers and Rational Numbers
- Lesson 2.2.4：Multiplication as Repeated Addition
- Lesson 2.2.5：Multiplication of Portions
- Lesson 2.2.6：Multiplying Mixed Numbers
- Lesson 2.3.1：Choosing a Scale and Graphing Data
- Lesson 2.3.2：More Graph Scaling

第3章 运算性质（Arithmetic Properties）

- 来源：Course 2 Text to CCSS Standards（PDF）
- Lesson 3.1.1：Grouping Expressions
- Lesson 3.1.2：Identifying Terms in Expressions
- Lesson 3.2.1：Subtraction of Integers
- Lesson 3.2.2：Connecting Addition and Subtraction
- Lesson 3.2.3：Multiplication as Repeated Subtraction
- Lesson 3.2.4：Multiplication with Decimals
- Lesson 3.2.5：Addition, Subtraction, Multiplication, and Division of Integers
- Lesson 3.3.1：Division with Rational Numbers

- Lesson 3.3.2：Division with Decimals
- Lesson 3.3.3：Arithmetic Properties

第4章 比例和代数式（Proportions & Expressions）

- 来源：Course 2 Text to CCSS Standards（PDF）
- Lesson 4.1.1：Similar Figures
- Lesson 4.1.2：Scale Drawings
- Lesson 4.2.1：Recognizing Proportional Relationships
- Lesson 4.2.2：Proportional Relationships with Tables and Graphs
- Lesson 4.2.3：Unit Rate and Proportional Equations
- Lesson 4.2.4：Connecting Representations of Proportional Relationships
- Lesson 4.3.1：Combining Like Terms
- Lesson 4.3.2：Distributive Property
- Lesson 4.3.3：Simplifying with Zero

第5章 概率和解应用题（Probability & Solving Word Problems）

- 来源：Course 2 Text to CCSS Standards（PDF）
- Lesson 5.1.1：Part-Whole Relationships
- Lesson 5.1.2：Finding and Using Percentages
- Lesson 5.2.1：Probability Games
- Lesson 5.2.2：Computer Simulations of Probability
- Lesson 5.2.3：Compound Independent Events
- Lesson 5.2.4：Probability Tables
- Lesson 5.2.5：Probability Trees
- Lesson 5.2.6：Compound Events
- Lesson 5.3.1：Describing Relationships Between Quantities
- Lesson 5.3.2：Solving a Word Problem
- Lesson 5.3.3：Strategies for Using the 5-D Process
- Lesson 5.3.4：Using Variables to Represent Quantities in Word Problems
- Lesson 5.3.5：More Word Problem Solving

第6章 解不等式和方程（Solving Inequalities & Equations）

- 来源：Course 2 Text to CCSS Standards（PDF）
- Lesson 6.1.1：Comparing Expressions
- Lesson 6.1.2：Comparing Quantities with Variables
- Lesson 6.1.3：One Variable Inequalities
- Lesson 6.1.4：Solving One Variable Inequalities
- Lesson 6.2.1：Solving Equations
- Lesson 6.2.2：Checking Solutions and the Distributive Property
- Lesson 6.2.3：Solving Equations and Recording Work
- Lesson 6.2.4：Using a Table to Write Equations from Word Problems
- Lesson 6.2.5：Writing and Solving Equations
- Lesson 6.2.6：Cases With Infinite or No Solutions

- Lesson 6.2.7：Choosing a Solving Strategy

第7章 比例和百分数（Proportions & Percents）

- 来源：Course 2 Text to CCSS Standards（PDF）
- Lesson 7.1.1：Distance, Rate, and Time
- Lesson 7.1.2：Scaling Quantities
- Lesson 7.1.3：Solving Problems Involving Percents
- Lesson 7.1.4：Equations with Fraction and Decimal Coefficients
- Lesson 7.1.5：Creating Integer Coefficients
- Lesson 7.1.6：Creating Integer Coefficients Efficiently
- Lesson 7.1.7：Percent Increase and Decrease
- Lesson 7.1.8：Simple Interest
- Lesson 7.2.1：Finding Missing Information in Proportional Relationships
- Lesson 7.2.2：Solving Proportions

第8章 统计和角的关系（Statistics & Angle Relationships）

- 来源：Course 2 Text to CCSS Standards（PDF）
- Lesson 8.1.1：Measurement Precision
- Lesson 8.1.2：Comparing Distributions
- Lesson 8.2.1：Representative Samples
- Lesson 8.2.2：Inference From Random Samples
- Lesson 8.3.1：Introduction to Angles
- Lesson 8.3.2：Classifying Angles
- Lesson 8.3.3：Constructing Shapes
- Lesson 8.3.4：Building Triangles

第9章 圆和体积（Circles & Volume）

- 来源：Course 2 Text to CCSS Standards（PDF）
- Lesson 9.1.1：Circumference, Diameter, and Pi
- Lesson 9.1.2：Area of Circles
- Lesson 9.1.3：Area of Composite Shapes
- Lesson 9.2.1：Surface Area and Volume
- Lesson 9.2.2：Cross Sections
- Lesson 9.2.3：Volume of a Prism
- Lesson 9.2.4：Volume of Non-Rectangular Prisms

Course 3（八年级） · 10 章 · 85 课

第1章 解题和正比例关系（Problem Solving & Proportional Relationships）

- 来源：Course 3 Text to CCSS Standards（PDF）
- Lesson 1.1.1：Interpreting Graphs
- Lesson 1.1.2：Finding and Generalizing Patterns
- Lesson 1.1.3：The Algebra Walk
- Lesson 1.1.4：Collecting, Organizing, and Analyzing Data
- Lesson 1.2.1：Proportional Relationships with Graphs and Tables
- Lesson 1.2.2：Strategies for Solving Proportional Relationships

第2章 用变量化简（Simplifying with Variables）

- 来源：Course 3 Text to CCSS Standards（PDF）
- Lesson 2.1.1：Exploring Variables and Expressions
- Lesson 2.1.2：Simplifying Expressions by Combining Like Terms
- Lesson 2.1.3：Writing Algebraic Expressions
- Lesson 2.1.4：Using Zero to Simplify Algebraic Expressions
- Lesson 2.1.5：Using Algebra Tiles to Simplify Algebraic Expressions
- Lesson 2.1.6：Using Algebra Tiles to Compare Expressions
- Lesson 2.1.7：Simplifying and Recording Work
- Lesson 2.1.8：Using Algebra Tiles to Solve for x
- Lesson 2.1.9：More Solving Equations

第3章 图像和方程（Graphs & Equations）

- 来源：Course 3 Text to CCSS Standards（PDF）
- Lesson 3.1.1：Extending Patterns and Finding Rules
- Lesson 3.1.2：Using Tables, Graphs, and Rules to Make Predictions
- Lesson 3.1.3：Using a Graphing Calculator and Identifying Solutions
- Lesson 3.1.4：Completing Tables and Drawing Graphs
- Lesson 3.1.5：Graphs, Tables, and Rules
- Lesson 3.1.6：Complete Graphs
- Lesson 3.1.7：Identifying Common Graphing Errors
- Lesson 3.2.1：Solving Equations and Checking Solutions
- Lesson 3.2.2：Determining the Number of Solutions
- Lesson 3.2.3：Problem Solving With Equations
- Lesson 3.2.4：More Solving Equations to Solve Problems
- Lesson 3.2.5：Distributive Property Equations

第4章 多种表示（Multiple Representations）

- 来源：Course 3 Text to CCSS Standards（PDF）
- Lesson 4.1.1：Finding Connections Between Representations

- Lesson 4.1.2：Seeing Growth in Different Representations
- Lesson 4.1.3：Connecting Linear Rules and Graphs
- Lesson 4.1.4：y = mx + b
- Lesson 4.1.5：Checking the Connections
- Lesson 4.1.6：Graphing a Line Without an x → y Table
- Lesson 4.1.7：Completing the Web

第5章 方程组（Systems of Equations）

- 来源：Course 3 Text to CCSS Standards（PDF）
- Lesson 5.1.1：Working with Multi-Variable Equations
- Lesson 5.1.2：Solving Equations with Fractions
- Lesson 5.2.1：Introduction to Systems of Equations
- Lesson 5.2.2：Writing Rules from Word Problems
- Lesson 5.2.3：Solving Systems Algebraically
- Lesson 5.2.4：Strategies for Solving Systems

第6章 变换和相似（Transformations & Similarity）

- 来源：Course 3 Text to CCSS Standards（PDF）
- Lesson 6.1.1：Rigid Transformations
- Lesson 6.1.2：Rigid Transformations on a Coordinate Graph
- Lesson 6.1.3：Describing Transformations
- Lesson 6.1.4：Using Rigid Transformations
- Lesson 6.2.1：Multiplication and Dilation
- Lesson 6.2.2：Dilations and Similar Figures
- Lesson 6.2.3：Identifying Similar Shapes
- Lesson 6.2.4：Similar Figures and Transformations
- Lesson 6.2.5：Working With Corresponding Sides
- Lesson 6.2.6：Solving Problems Involving Similar Shapes

第7章 斜率和相关（Slope & Association）

- 来源：Course 3 Text to CCSS Standards（PDF）
- Lesson 7.1.1：Circle Graphs
- Lesson 7.1.2：Organizing Data in a Scatterplot
- Lesson 7.1.3：Identifying and Describing Association
- Lesson 7.2.1：y = mx + b Revisited
- Lesson 7.2.2：Slope
- Lesson 7.2.3：Slope in Different Representations
- Lesson 7.2.4：More About Slope
- Lesson 7.2.5：Proportional Equations
- Lesson 7.3.1：Using Equations to Make Predictions
- Lesson 7.3.2：Describing Association Fully
- Lesson 7.3.3：Association Between Categorical Variables

第8章 指数和函数（Exponents & Functions）

- 来源：Course 3 Text to CCSS Standards（PDF）
- Lesson 8.1.1：Patterns of Growth in Tables and Graphs
- Lesson 8.1.2：Compound Interest
- Lesson 8.1.3：Linear and Exponential Growth
- Lesson 8.2.1：Exponents and Scientific Notation
- Lesson 8.2.2：Exponent Rules
- Lesson 8.2.3：Negative Exponents
- Lesson 8.2.4：Operations with Scientific Notation
- Lesson 8.3.1：Functions in Graphs and Tables

第9章 角和勾股定理（Angles & Pythagorean Theorem）

- 来源：Course 3 Text to CCSS Standards（PDF）
- Lesson 9.1.1：Parallel Line Angle Pair Relationships
- Lesson 9.1.2：Finding Unknown Angles in Triangles
- Lesson 9.1.3：Exterior Angles in Triangles
- Lesson 9.1.4：AA Triangle Similarity
- Lesson 9.2.1：Side Lengths and Triangles
- Lesson 9.2.2：Pythagorean Theorem
- Lesson 9.2.3：Understanding Square Root
- Lesson 9.2.4：Real Numbers
- Lesson 9.2.5：Applications of Pythagorean Theorem
- Lesson 9.2.6：Pythagorean Theorem in Three Dimensions
- Lesson 9.2.7：Pythagorean Theorem Proofs

第10章 表面积和体积（Surface Area & Volume）

- 来源：Course 3 Text to CCSS Standards（PDF）
- Lesson 10.1.1：Cube Roots
- Lesson 10.1.2：Surface Area and Volume of a Cylinder
- Lesson 10.1.3：Volumes of Cones and Pyramids
- Lesson 10.1.4：Volume of a Sphere
- Lesson 10.1.5：Applications of Volume

说明和待核事项

- CPM 官网目前同时有 Core Connections 第 2 版（Parent Guide）和第 3 版（Guided Skill Support）的家长资料。本大纲的课名来自《Text to CCSS Standards》对照表，这份 PDF 没有写明对应第几版，课名可能和最新版有出入，标「待核」。
- PDF 里少数课名跨两行，本项目把两行拼成一个标题（例如 Course 3 的 10.1.2 Surface Area and Volume of a Cylinder），拼接处待核。
- 本大纲和海报由 AI 辅助整理，**公开发布或用于教学前需要人工审核**。